

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 9 - Summative Assessment 1

Summative Assessment (SA-1)

Class: Class 9

Assessment Type: Summative Assessment (SA-1)

Total Marks: 40

Duration: 2 hours

Topics Covered: Python Fundamentals, Data Structures, SQL, Web Technologies

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Part A: MCQ (10 x 1 = 10 marks)

a. What is the output?

```
x = [1, 2, 3, 4, 5]
print(x[::2] + x[1::2])
```

- a. [1, 3, 5, 2, 4]
- b. [1, 2, 3, 4, 5]
- c. [1, 3, 5, 1, 3, 5]
- d. Error

b. Which of the following is the correct way to create a dictionary?

- c. dict = {"a": 1, "b": 2}
- d. dict = [("a", 1), ("b", 2)]
- e. dict = [{"a"}: 1]
- f. dict = dict("a"=1)

c. What will be the output?

```
s = "Python"
print(s[-3:])
```

- a. "tho"
- b. "hon"
- c. "tho"
- d. "no"

d. Which SQL command removes a table completely from the database?

- e. DELETE TABLE
- f. REMOVE TABLE
- g. DROP TABLE
- h. TRUNCATE TABLE

e. Which HTML tag is used to define an unordered list?

- f.
- g.

- h.
- i. <list>
- f. What is the time complexity of searching an element in a sorted list using binary search?
- g. $O(n)$
- h. $O(\log n)$
- i. $O(n^2)$
- j. $O(1)$
- g. Which CSS property is used to change the font family?
- h. text-font
- i. font-family
- j. font-style
- k. text-family
- h. What will be the output?
- ```
d = {"x": 1, "y": 2, "z": 3}
print(list(d.values()))
```
- a. ["x", "y", "z"]
- b. [1, 2, 3]
- c. ("x", "y", "z")
- d. Error
- i. Which JavaScript method selects an element by its ID?
- j. document.querySelector()
- k. document.getElementById()
- l. document.getElement()
- m. document.findById()
- j. What does COUNT(\*) do in SQL?
- k. Counts only non-NULL values
- l. Counts all rows including NULL values
- m. Counts only distinct values
- n. Error

## Part B: Short Answer (8 x 2 = 16 marks)

- a. Write a Python function merge\_dicts(d1, d2) that merges two dictionaries. If a key exists in both, the values should be added together. Test with {"a": 1, "b": 2} and {"b": 3, "c": 4}.
- b. Write the output of:

```
lst = [[1, 2, 3], [4, 5, 6], [7, 8, 9]]
result = [lst[i][i] for i in range(len(lst))]
print(result)
```

Explain the logic.

- c. Write SQL queries for a table Employees(emp\_id, name, department, salary, joining\_date):
- d. Find employees who joined in the last 2 years
- e. Find the second highest salary
- f. Find departments where average salary exceeds 50000
- d. Differentiate between DELETE, TRUNCATE, and DROP with one example each.
- e. Explain the CSS **Flexbox** layout model. Write CSS to create a three-column layout where the middle column is twice the width of the side columns.
- f. What is **responsive web design**? Explain three techniques used to make a web page responsive.
- g. Write a Python program to implement a **stack** using a list with operations: push, pop, peek, is\_empty, and display.
- h. Write a SQL query to find the **nth highest salary** from an Employees table. Write it for n = 5.

### Part C: Long Answer (4 x 3 = 12 marks)

- a. Write a complete Python program that implements a **Library Management System** using dictionaries and lists. The program should:

- Store book details: title, author, ISBN, available copies
- Allow operations: add book, issue book, return book, search by title/author
- Display all issued books and their due dates
- Calculate and display fines for overdue books (₹5 per day)
- Use functions for each operation with proper error handling

- b. Write SQL queries for an **E-Commerce Database** with tables:

```
Customers(cust_id, name, email, city)
Products(prod_id, name, price, category, stock)
Orders(order_id, cust_id, order_date, total_amount)
Order_Items(order_id, prod_id, quantity, price)
```

Write queries to:

- a. Find the top 5 customers by total spending
- b. Find products that have never been ordered
- c. Calculate monthly revenue for the current year
- d. Find the most popular product category
- e. List customers who ordered more than 3 different products
- c. Create a complete web page for a **Student Report Card Generator** that:
  - Has a form to enter: Student Name, Class, Section, Roll Number, and marks for 5 subjects
  - Uses CSS Grid layout for the form
  - JavaScript to:
    - Validate all inputs (marks must be between 0 and 100)
    - Calculate total, percentage, and grade
    - Generate a styled report card below the form
    - Include a print button that opens the print dialog

d. Write a Python program that implements a **Contact Book** using:

- File handling to persist data between runs
- Dictionary of lists to store contacts
- Operations: add, search, update, delete, display all contacts
- Search by name (partial match), phone number, or email
- Sort contacts alphabetically or by creation date
- Export contacts to a CSV file

### **Part D: Application Based (2 x 1 = 2 marks)**

a. A school wants to create a system that tracks student attendance. Explain how:

- Python data structures (dictionaries, lists) can store daily attendance
- SQL database can store historical attendance data with proper schema
- A web interface (HTML/CSS/JS) can display attendance reports
- Include the key components needed for each technology

b. An online bookstore needs an inventory system. Analyse how:

- SQL can maintain the book catalog with proper relationships
- Python can process bulk imports from CSV files
- A web form can allow adding new books
- Consider at least one SQL query that would be commonly used